



9695 - Unveiling the hidden nature of highly-ionizing objects at cosmic dawn

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

Folder	Observation	Label	Observing Template	Science Target
NIRSpec R1000 on GLASS-JWST				
	1	PA 35 - GHZ2 + GHZ9 + GHZ7 + FILLERS	NIRSpec MultiObject Spectroscopy	(14) A2744_NIRSpec_R1000_V3
	2	PA 200 - GHZ2 + GHZ9 + GHZ1 + FILLERS	NIRSpec MultiObject Spectroscopy	(14) A2744_NIRSpec_R1000_V3
	3	PA 205 - GHZ2 + GHZ9 + GHZ4 + FILLERS	NIRSpec MultiObject Spectroscopy	(14) A2744_NIRSpec_R1000_V3

ABSTRACT

The present proposal will characterize two outstanding objects at $z > 10$ that have the potential to shed light on the connection among three puzzling JWST discoveries: 1) the high abundance of bright galaxies at $z > 9$, 2) the large fraction of high-redshift AGN, and 3) the existence of a class of highly-ionizing, N-enhanced objects. The two sources, GHZ2/GLASS-z12 ($z=12.34$) and GHZ9 ($z=10.145$) show UV emission lines whose strength is unprecedented among $z > 10$ objects, in particular a prominent CIV emission ($EW > 40\text{\AA}$), and detection of the NIV and NIII lines implying a super-solar N/O abundance. GHZ9 is the most distant X-ray AGN candidate ever found, with an X-ray luminosity compatible with a $M_{\text{BH}} \sim 10^8 M_{\text{sun}}$. Line ratio diagnostics suggest that these may be composite objects, with both an AGN and compact star-formation contributing to their spectra. The proposed observations have the goal and sensitivity to detect even a minor AGN contribution to the observed emission to unveil the nature of these unique sources. NIRSPEC G235M/F170LP medium resolution spectra will detect or put firm upper limits ($< 20\%$ of the total flux) on broad ($\text{FWHM} > 500 \text{ km/s}$) CIV components, and will confirm that they are N-enhanced through a robust detection of the NIII line. The large number of high-redshift sources confirmed in the GLASS-JWST field will also enable a robust investigation of 2 objects at $z \sim 10.4\text{--}10.7$ with CIII $EW > 20\text{\AA}$ compatible with AGN emission, and of the bright galaxy GHZ1/GLASS-z10 at $z=9.88$ showing a detection of the NIII-line.

OBSERVING DESCRIPTION

The program aims at assessing whether GHZ9 and GHZ2/GLASS-z12 are dominated by star-formation or AGN emission, and confirming they are nitrogen-enhanced objects.

We will target the GLASS-JWST NIRCam field with the G235M/F170LP configuration that covers the UV rest frame range where the most constraining features of the sources are found. The observations will assess the presence of broad (>500 Km/s) components of the CIV1549 line down to a 20% of the total flux, and confirm the detection of the NIII1750 multiplet to firmly establish the nitrogen enhancement in combination with NIV1488 and OIII1663. The medium resolution spectrum will robustly measure the flux of the HeII1640 line which is partially blended with OIII1663 in the prism. The MSA will also obtain deep spectra of other three other confirmed $z>9.5$ galaxies: GHZ4 ($z=10.66$), GHZ7 ($z=10.43$), which show high excitation lines compatible with AGN emission, and the bright galaxy GHZ1/GLASS-z10 ($z=9.88$) showing NIII detection in the PRISM.

The program adopts the readout pattern NRSIRS2, midpoint centering, standard 3-shutter "slits", a 3-point nodding and a 80-shutter dither along the dispersion direction to bridge the detector gap. The observations are designed with APT 2025.5.3 to obtain the best spectral coverage for the two main targets by avoiding both blue and red cutoffs, and forcing the entire wavelength range to be observed at both dither positions to maximise the exposure time on all emission lines.

The desired sensitivity is reached with a total on-source time of 32.9 hours. The total charged time is 39 hours including overheads.

Our requested exposure time cannot be acquired in one observation due to data volume constraints, we thus divided the program in three observations and adjusted centering and position angle of the pointings to maximise the coverage of other sources at $z>9.5$. Each observation comprises 6 exposures (3-nod x 2-dither positions). The requested on-source time is reached with two observations of 5 integrations of 14 groups per exposure, and a third observation of 10 integrations of 13 groups per exposure. The relevant Aperture PA ranges needs to be restricted in order to observe the two primary targets with continuous spectral coverage with no gaps or cutoffs in each pointing, while obtaining the requested coverage of the other three confirmed $z>9.5$ galaxies: pointing one (on GHZ2, GHZ9 and GHZ7) must be fixed at PA=35, pointing two (on GHZ2, GHZ9 and GHZ1) at PA=200, and pointing three (on GHZ2, GHZ9 and GHZ4) at PA=205.

Remaining slits will be allocated to filler targets selected from the ultra-deep NIRCam imaging available on the GLASS-JWST ERS field.

Proposal 9695 - Targets - Unveiling the hidden nature of highly-ionizing objects at cosmic dawn

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(14)	A2744_NIRSpec_R1000_V3	RA: 00 14 0.9665 (3.5040271d) Dec: -30 20 49.25 (-30.34701d) Equinox: J2000		
	Comments: Description=[]				

Proposal 9695 - Observation 1 - Unveiling the hidden nature of highly-ionizing objects at cosmic dawn

Observation	Proposal 9695, Observation 1: PA 35 - GHZ2 + GHZ9 + GHZ7 + FILLERS										Fri Mar 13 17:04:51 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
Diagnostics	(PA 35 - GHZ2 + GHZ9 + GHZ7 + FILLERS (Obs 1)) Warning (Form): Config c1 (#1) has 4 filler slits affected by failed closed shutters.										
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(14)	A2744_NIRSpec_R1000_V3	RA: 00 14 0.9665 (3.5040271d) Dec: -30 20 49.25 (-30.34701d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List		Filler Candidate List		Spectral Overlap Map	Spectral Overlap Threshold
	MSATA	false		No	MSA Center	GHZ2 and GHZ9 (2 sources)		other GHZs + FILLERS (10956 sources)		jwst-nirspec-g235m	1.5
Reference Stars											
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G235M/F170LP)	c1	3 Shutter Slitlet	3.4713075 Degrees - 30.31756666666 664 Degrees	35.016603447281 88			3	15	15537.168
	2	1 (G235M/F170LP)	c2	3 Shutter Slitlet	3.4656671666666 67 Degrees - 30.31415611111 117 Degrees	35.019462329373 35			3	15	15537.168

Proposal 9695 - Observation 1 - Unveiling the hidden nature of highly-ionizing objects at cosmic dawn

Special Requirements	Aperture PA Range 35 to 35 Degrees (V3 256.4254303 to 256.4254303) MSA Planned Aperture PA 35.0000 to 35.0000 Degrees (V3 256.4254303 to 256.4254303)
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Proposal 9695 - Observation 2 - Unveiling the hidden nature of highly-ionizing objects at cosmic dawn

Observation	Proposal 9695, Observation 2: PA 200 - GHZ2 + GHZ9 + GHZ1 + FILLERS										Fri Mar 13 17:04:51 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(14)	A2744_NIRSpec_R1000_V3	RA: 00 14 0.9665 (3.5040271d) Dec: -30 20 49.25 (-30.34701d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold			
	MSATA	false	No	MSA Center	ALL GHZs (6 sources)	other GHZs + FILLERS (10956 sources)	jwst-nirspec-g235m	1.5			
Reference Stars											
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G235M/F170LP)	c1	3 Shutter Slitlet	3.5084858333333 333 Degrees - 30.34154722222 232 Degrees	199.99774121657 526			3	15	15537.168
	2	1 (G235M/F170LP)	c2	3 Shutter Slitlet	3.5149586250000 002 Degrees - 30.34357972222 216 Degrees	199.99448057051 61			3	15	15537.168

Proposal 9695 - Observation 2 - Unveiling the hidden nature of highly-ionizing objects at cosmic dawn

Special Requirements	Aperture PA Range 200 to 200 Degrees (V3 61.4254303 to 61.4254303) MSA Planned Aperture PA 200.0000 to 200.0000 Degrees (V3 61.4254303 to 61.4254303)
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Proposal 9695 - Observation 3 - Unveiling the hidden nature of highly-ionizing objects at cosmic dawn

Observation	Proposal 9695, Observation 3: PA 205 - GHZ2 + GHZ9 + GHZ4 + FILLERS										Fri Mar 13 17:04:51 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
Diagnostics	(PA 205 - GHZ2 + GHZ9 + GHZ4 + FILLERS (Obs 3)) Warning (Form): Config c1 (#1) has 3 filler slits affected by failed closed shutters.										
	(PA 205 - GHZ2 + GHZ9 + GHZ4 + FILLERS (Obs 3)) Warning (Form): Config c2 (#2) has 1 filler slits affected by failed closed shutters.										
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(14)	A2744_NIRSpec_R1000_V3	RA: 00 14 0.9665 (3.5040271d) Dec: -30 20 49.25 (-30.34701d) Equinox: J2000								
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List		Filler Candidate List		Spectral Overlap Map	Spectral Overlap Threshold
	MSATA	false		No	MSA Center	ALL GHZs (6 sources)		other GHZs + FILLERS (10956 sources)		jwst-nirspec-g235m	1.5
Reference Stars											
Spectral Elements	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (G235M/F170LP)	c1	3 Shutter Slitlet	3.5068562500000 002 Degrees - 30.341883333333 328 Degrees	204.99856289153 942			3	30	28886.002
	2	1 (G235M/F170LP)	c2	3 Shutter Slitlet	3.5130990833333 335 Degrees - 30.344395277777 778 Degrees	204.99541830697 325			3	30	28886.002

Proposal 9695 - Observation 3 - Unveiling the hidden nature of highly-ionizing objects at cosmic dawn

Special Requirements	Aperture PA Range 205 to 205 Degrees (V3 66.4254303 to 66.4254303) MSA Planned Aperture PA 205.0000 to 205.0000 Degrees (V3 66.4254303 to 66.4254303)
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